

class17_PCA

Core Unix commands

Most unix commands have super short names, which makes them quick to type but annoying to learn. Major file system related commands include.

pwd: Print working directory ls: List files and directories cd: Change Directories mkdir: Make directories rm: remove files and directories (delete) cp: copy files (source > destination) mv: Move files (basically re-name) nano: A wee text editor (very basic but always available)

curl: Download files wget: Download files from the web tar -zxvf:UnTar (unpackage) Tar archive files gunzip : UnZip files \$PATH: The places (dirs) to look for programs

AWS EC2 Instance

Connect to my instance with: `ssh -i ~/Downloads/bimm143_eloise.pem ubuntu@ec2-54-202-63-98.us-west-2.compute.amazonaws.com`

Github link: https://github.com/EloiseSimpson/Bimm143_github/tree/main

Secure copy files between machines, in this case from our instance to our laptop `scp -i ~/Downloads/bimm143_eloise.pem ubuntu@ec2-54-202-63-98.us-west-2.compute.amazonaws.com:/home/ubuntu/work/ .`

Class 17 instance

`ssh -i "bimm143_eloise.pem" ubuntu@ec2-52-33-145-234.us-west-2.compute.amazonaws.com`

`export KEY=~/.Downloads/bimm143_eloise.pem export SERVER=ubuntu@ec2-52-33-145-234.us-west-2.compute.amazonaws.com`

`ssh -i $KEY $SERVER scp -r`

Downstream Analysis

```
library(tximport)

base_path <- "class17"
folders <- dir(path = base_path, pattern = "SRR21568*", full.names = TRUE)
samples <- sub("_quant$", "", basename(folders))
files <- file.path(folders, "abundance.h5")
names(files) <- samples

txi.kallisto <- tximport(files, type = "kallisto", txOut = TRUE)
```

1 2 3 4

```
head(txi.kallisto$counts)
```

	SRR2156848	SRR2156849	SRR2156850	SRR2156851
ENST00000539570	0	0	0.00000	0
ENST00000576455	0	0	2.62037	0
ENST00000510508	0	0	0.00000	0
ENST00000474471	0	1	1.00000	0
ENST00000381700	0	0	0.00000	0
ENST00000445946	0	0	0.00000	0

```
colSums(txi.kallisto$counts)
```

SRR2156848	SRR2156849	SRR2156850	SRR2156851
2563611	2600800	2372309	2111474

```
sum(rowSums(txi.kallisto$counts)>0)
```

[1] 94561

```
to.keep <- rowSums(txi.kallisto$counts) > 0
kset.nonzero <- txi.kallisto$counts[to.keep,]
```

```
keep2 <- apply(kset.nonzero,1,sd)>0
x <- kset.nonzero[keep2,]
```

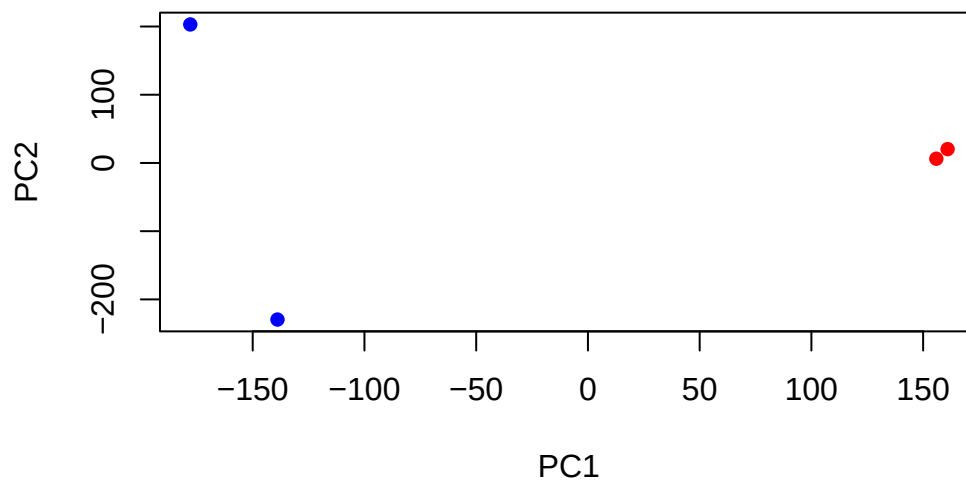
Principal Component Analysis

```
pca <- prcomp(t(x), scale=TRUE)
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4
Standard deviation	183.6379	177.3605	171.3020	1e+00
Proportion of Variance	0.3568	0.3328	0.3104	1e-05
Cumulative Proportion	0.3568	0.6895	1.0000	1e+00

```
plot(pca$x[,1], pca$x[,2],
     col=c("blue","blue","red","red"),
     xlab="PC1", ylab="PC2", pch=16)
```



```

library(ggplot2)
library(ggrepel)

mycols <- c("blue","blue","red","red")

ggplot(pca$x) +
  aes(PC1, PC2, label=rownames(pca$x)) +
  geom_point( col=mycols ) +
  geom_text_repel( col=mycols ) +
  theme_bw()

```

